

parameter in `fsolve()` is the initial guess of the solution.

Geometric problem solving

With the power in hand, why not solve more complex geometric problems? Last time, we found the point of intersection of two straight lines. How about the intersection of a straight line and a circle? Let us have the following straight line and circle, defined in the Cartesian coordinate system, i.e., the x-y system:

$$4x + 3y = 24$$

$$x^2 + y^2 = 25$$

To be able to solve it using `fsolve`, let's consider the different variables 'x' and 'y' as fields of a vector X, say 'x' as X(1), 'y' as X(2). Then, the equations can be re-written as follows:

$$4 * X(1) + 3 * X(2) = 24$$

$$X(1)^2 + X(2)^2 = 25$$

...and hence could be solved using `fsolve` as follows:

```
$ octave -qf
octave:1> function Y = F(X)
> Y(1) = 4 * X(1) + 3 * X(2) - 24;
> Y(2) = X(1)^2 + X(2)^2 - 25;
> endfunction
octave:2> [Y, Fval, info] = fsolve(@F, [0; 0])
warning: matrix singular to machine precision, rcond = 0
warning: attempting to find minimum norm solution
warning: dgelsd: rank deficient 2x2 matrix, rank = 1
Y =

    3.0000
    4.0000

Fval =

    0.0000e+00    2.6691e-07

info = 1
octave:3>
```

So (3, 4) is the point of intersection. This can be verified by re-substituting the values in the original equations.

Try and solve these

Equipped with this knowledge, here are a couple of teasers for your brain:

1. Find three numbers, the product of which is 60, the sum of their squares is 50, and their sum is 12.
2. A sage came to a temple with some flowers and dipped all of them into the first magical pond of the temple and

got those back, squared. Then, he offered some of those flowers in the temple and dipped the remaining flowers into the second magical pond to get those back, doubled. Then, he again offered the same number of flowers, as offered earlier, and dipped the remaining flowers into the third magical pond to get those back, tripled, which he took back with him as *prasadam*. Now, the number of flowers he took back with him, is same as in each one of his offerings. Also, what he took back with him is thrice the number of flowers he came with to the temple. How many flowers did he come to the temple with?

If you think you have got the Octave code to solve the above questions, you may email the solution to me at the mail ID mentioned below. Going forward, we will start to specifically explore polynomials. 

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